Royal University of Phnom Penh
Faculty of Engineering
Major: Data Science and Engineering

EXPLAINABLE AI FOR DIABETES RISK PREDICTION AND ACTIONABLE INSIGHTS

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OUTLINE

- Background and Motivation
- Research Objective
- Literature Review
- Methodology
- Results
- Conclusion

Introduction

BACKGROUND AND MOTIVATION

- Diabetes is a chronic disease associated with serious complications such as heart disease, kidney failure, and vision loss
- Early risk prediction can support prevention and screening
- Machine learning identifies risk patterns from health and lifestyle survey data
- Explainable AI helps users understand model predictions, reveals reasoning behind model predictions

Introduction

PROBLEM STATEMENT

01 Traditional diabetes screening may require time, cost, and medical resources

02 Existing prediction systems often focus only on prediction accuracy without explaining the results clearly

03 Cambodia has limited AI-based diabetes prediction research using local health data

04 Health datasets may contain missing values and imbalanced data, which can affect model performance

05 Need for explainable risk prediction system to raise awareness and support early screening

Introduction

RESEARCH OBJECTIVES

- Develop machine learning models for diabetes risk prediction using survey-based health indicators
- Train and validate the models using BRFSS 2017 data, and evaluate the final model on Cambodia STEPS as an independent test set
- Compare model performance using Recall, Precision, Accuracy, and F1-score
- Apply Explainable AI (SHAP) to identify key diabetes risk factors and explain individual predictions
- Generate actionable insights to support diabetes risk awareness and early screening

Introduction

RESEARCH QUESTIONS



Can health survey data from Cambodia STEPS and BRFSS be used to predict diabetes risk without blood test data?



Which machine learning model is the best for finding people who may have diabetes risk?



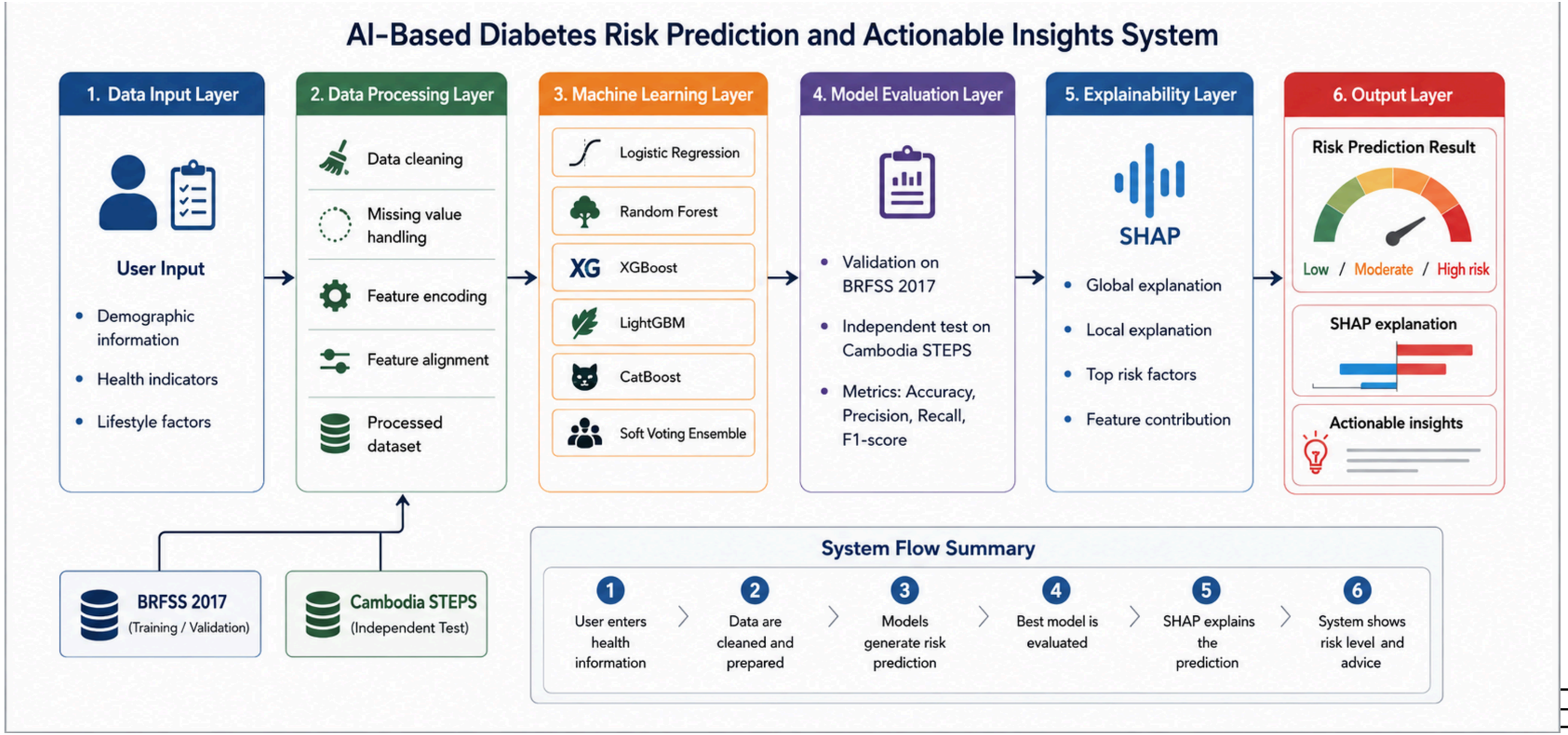
Which health indicators contribute most to diabetes risk prediction according to SHAP analysis?

LITERATURE REVIEW

Paper / Author	Model Used	Dataset Type	Accuracy / Result
Xie et al. (2019) — *Building Risk Prediction Models for Type 2 Diabetes Using Machine Learning Techniques*	SVM, Decision Tree, Logistic Regression, Random Forest, Neural Network, Naive Bayes	BRFSS 2014 survey data	Neural Network: **82.4% accuracy**, AUC **0.7949**. Decision Tree had highest sensitivity **51.6%**
Khokhar et al. — *Towards Transparent and Accurate Diabetes Prediction Using ML and XAI*	Random Forest, XGBoost, LightGBM, Ensemble Model + SHAP/LIME	Diabetes Binary Health Indicators dataset from BRFSS 2015	Ensemble model: **92.50% accuracy**, ROC-AUC **0.975**. Important features: BMI, Age, General Health, Income, Physical Activity
Katiyar et al. (2024) *Recent Advancements Using ML & DL Approaches for Diabetes Detection*	ML and Deep Learning Models	Multiple Diabetes Datasets	ML and DL models achieved high prediction performance, dataset quality significantly affects results
Li et al. (2025) Performance of Diabetes Risk Prediction Models	Statistical Models + Machine Learning Models	Population Health Datasets	Risk prediction models can support early screening and prevention of Type 2 Diabetes
Khokhar et al. (2025) Advances in Artificial Intelligence for Diabetes Prediction	Logistic Regression, SVM, Random Forest, XGBoost, CNN, RNN	Multiple Public Diabetes Datasets	Reviewed 53 studies, highlighted strong performance of ML models and the need for explainable and clinically relevant systems

Methodology

System Architecture



Methodology

Overall Framework



Methodology

DATASET OVERVIEW

Cambodia STEPS

Local survey data collected in Cambodia.

- Demographic indicators (age, sex, education)
- Health indicators (BMI, blood pressure, cholesterol)
- Lifestyle indicators (smoking, physical activity, diet)
- Use for independent testing
- Target: Diabetes status (0 = non-diabetes, 1 = diabetes)

BRFSS

Larger benchmark dataset from the United States.

- Similar health and lifestyle indicators
- Used for training 2017)
- Target: Diabetes status (0 = non-diabetes, 1 = diabetes)

SELECTED FEATURES



Demographic

Age
Sex
Education



Health Status

High Blood Pressure
High Cholesterol
BMI



Lifestyle & Healthcare Access

Smoker
Physical Activity
Fruits
Vegetables
Heavy Alcohol Consumption

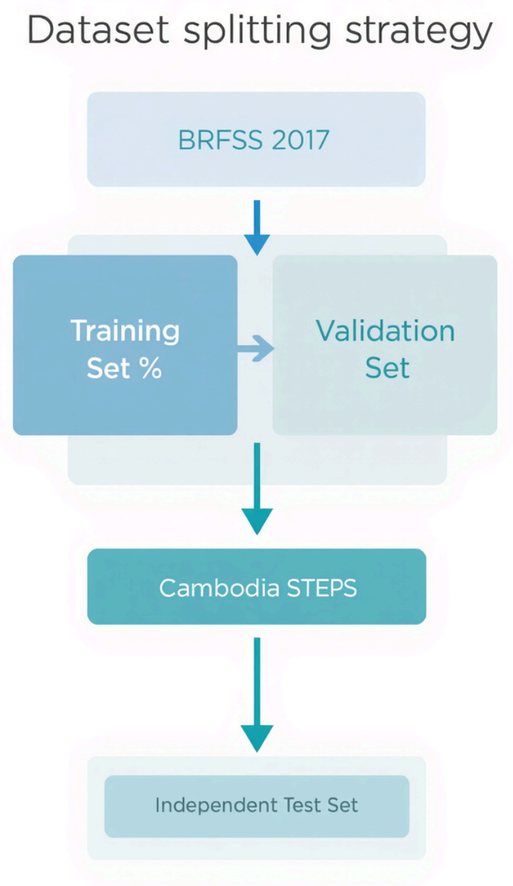
DATA SPLITTING STRATEGY

BRFSS 2017 Dataset

- 80% Training Set: Used for model learning and fitting
- 20% Validation Set: Used for model selection and decision threshold tuning

Cambodia STEPS Dataset

- Independent Test Set: Used for final cross-population evaluation only

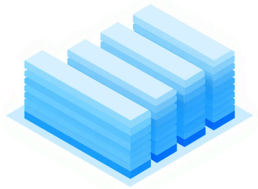


Methodology

MACHINE LEARNING MODELS

1. Logistic Regression
2. Random Forest
3. XGBoost
4. LightGBM
5. CatBoost
6. Soft Voting Ensemble





EVALUATION METRICS

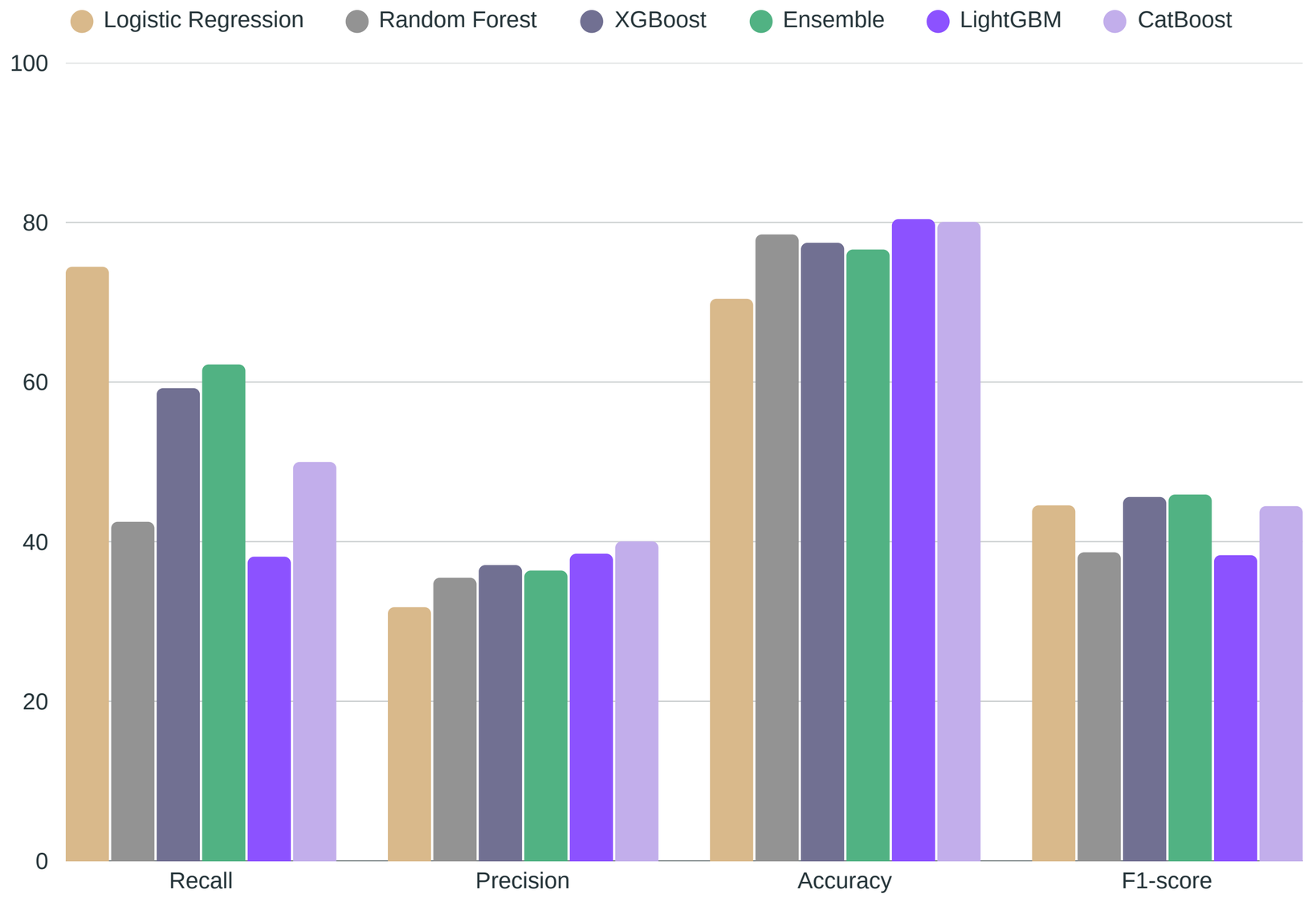
Recall

Precision

Accuracy

F1 Score

Model Evaluation and Results



Results

CAMBODIA STEPS INDEPENDENT TEST RESULT

- Final selected model: Soft Voting Ensemble
- Lower performance is expected because the datasets come from different populations and survey systems

This result is for risk awareness and early screening support, not clinical diagnosis.

```
============================================================
  INDEPENDENT TEST — Cambodia STEPS (once)
============================================================
  Model     : Soft Voting
  Threshold : 0.43
  Train data: BRFSS 2017 only
  Features  : 11 (['Sex', 'Age', 'Education', 'HighBP',

  Recall       0.7623 <- clinical priority
  Precision    0.2058
  Accuracy     0.5556
  F1           0.3240 <- selection metric

              precision    recall  f1-score   support

 No Diabetes       0.93      0.52      0.67       751
    Diabetes       0.21      0.76      0.32       122

    accuracy                           0.56       873
   macro avg       0.57      0.64      0.50       873
weighted avg       0.83      0.56      0.62       873
```

	Validation (BRFSS 2017)	Test (STEPS)
Recall	0.7046	0.7623
F1	0.4515	0.3240
Accuracy	0.7269	0.5556
Precision	0.3322	0.2058

EXPLAINABLE AI

Global Explanation

Overall feature importance across the entire dataset using SHAP summary plots.

- Identifies which features most influence diabetes risk predictions globally
- BMI, age, blood pressure, and general health rank as top contributors
- SHAP summary plot visualizes direction and magnitude of each feature's impact
- Enables model-level transparency for researchers and practitioners

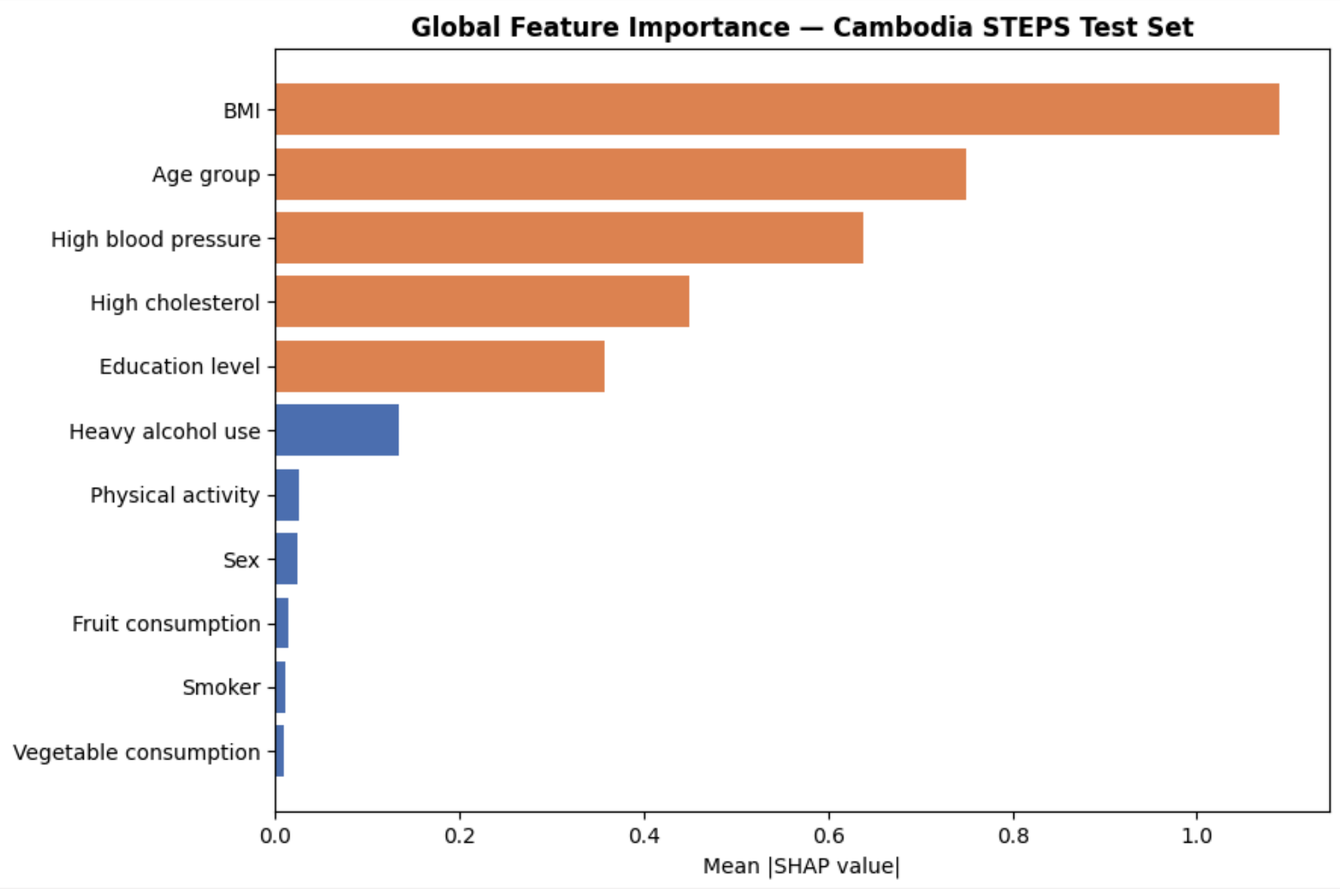
Local Explanation

Individual prediction rationale for a single patient using SHAP force plots.

- Explains why the model assigned a specific risk probability to one person
- Shows which features pushed the prediction higher or lower
- SHAP force plot highlights positive and negative contributing factors
- Builds trust by making individual predictions interpretable and actionable

SHAP ANALYSIS

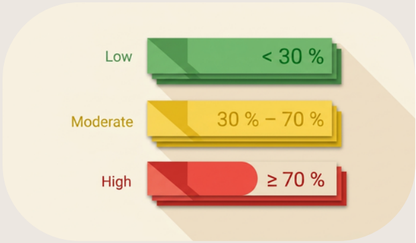
SHAP was used to explain how the final Soft Voting Ensemble model makes diabetes risk predictions



Results

RISK OUTPUT & ACTIONABLE INSIGHTS

Risk Levels



🟢 Low Risk
Probability < 30%
Continue healthy habits and routine check-ups.
🟡 Moderate Risk
Probability 30% – 69%
Consult a healthcare provider and monitor key indicators.
🔴 High Risk
Probability ≥ 70%
Seek medical advice promptly and take preventive action.

Actionable Insights

Actionable Insights:

- Check blood sugar with a healthcare provider.
- Maintain a healthy BMI.
- Increase physical activity.
- Improve diet by reducing sugar and eating more vegetables and fruits.
- Monitor blood pressure and cholesterol regularly.

Results

FINAL OUTPUT EXAMPLE

Patient Result

78%

High Risk



Predicted diabetes risk probability

Main Reasons

1. BMI – biggest reason increasing risk
2. Age – second strongest factor
3. High blood pressure – increases risk
4. Physical activity – helps reduce risk

Group Insight



- Health Status: 65%
- Lifestyle & Healthcare Access: 25%
- Demographic: 10%

Health Status is the main category driving this patient's risk

Actionable Recommendation

1. Increase regular physical activity
2. Monitor and reduce BMI
3. Check blood pressure regularly
4. Consult a healthcare professional

For education and early awareness only not a medical diagnosis

This final output extends binary prediction into probability, risk level, explanation, and action

CONCLUSION & FUTURE WORK

Conclusion:

- Built an explainable machine learning framework for diabetes risk prediction
- Trained and validated models using BRFSS 2017 data
- Tested the final selected model on Cambodia STEPS
- Applied SHAP to explain global and individual predictions
- Generated risk levels and actionable insights for early awareness

Future Work:

- Collect more Cambodia local health data
- Add clinical data such as glucose or HbA1c if available
- Test more threshold strategies to improve all the metric results
- Deploy and evaluate the web demo with healthcare feedback



Thank You